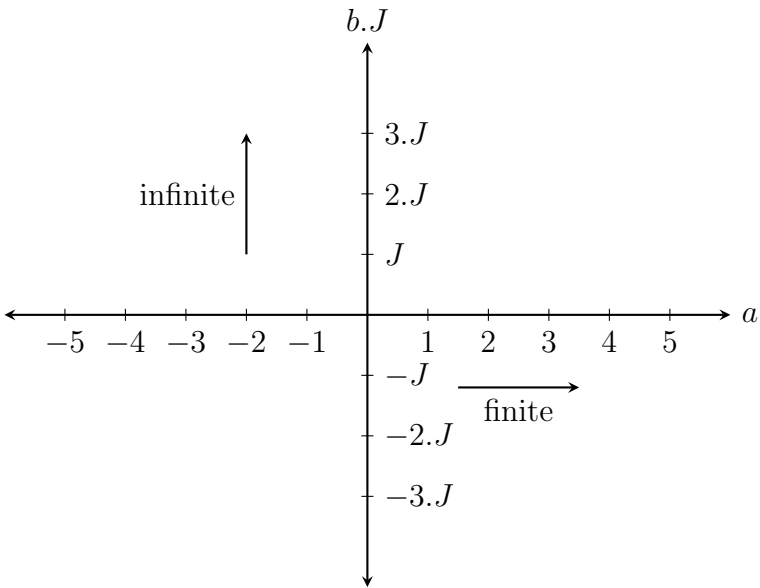


THE J-MATH

Let $\frac{1}{0} = J$

Let there be a 2D space, such that:



It is expressed in the form of

$$“a + bJ” \rightarrow \text{J-numbers}$$

$$\{a, b \in \mathbb{R}\}$$

$$\begin{aligned} a &= \text{finite part} \\ b.J &= \text{infinite part} \end{aligned}$$

$0 \cdot (a + bJ) = b$

Addition

Addition is performed **component-wise**. Let there be two J-numbers, x and y , such that:

$$\begin{aligned}x &= a + bJ \\ y &= c + dJ\end{aligned}$$

Then, the sum is:

$$\begin{aligned}x + y &= (a + bJ) + (c + dJ) \\ x + y &= (a + c) + (b + d)J\end{aligned}$$

Exponential/Power Rules

Since $J = \frac{1}{0}$, we establish the following rules:

1. For $n > 0$:

$$J^n = \left(\frac{1}{0}\right)^n = J = \frac{1}{0}$$

2. For $n = 0$:

$$J^0 = \left(\frac{1}{0}\right)^0 = 1$$

3. For $n < 0$:

$$J^n = \left(\frac{1}{0}\right)^n = 0$$

Multiplication [Piecewise]

Let there be two J-numbers, x and y , such that:

$$\begin{aligned}x &= a + bJ \\ y &= c + dJ\end{aligned}$$

Then, the product $x \cdot y$ is defined as:

$$x \cdot y = \begin{cases} b & \text{if } y = 0 \quad [\text{i.e., } c = 0 \text{ and } d = 0] \\ d & \text{if } x = 0 \quad [\text{i.e., } a = 0 \text{ and } b = 0] \\ ac + (ad + bc + bd)J & \text{otherwise} \end{cases}$$

Standard Real Numbers as J-Numbers

All standard real numbers can be written as J-numbers, such that their J-coefficient (b) is zero.

$$\begin{aligned}0 &= 0 + 0J \\ 7 &= 7 + 0J \\ -5 &= -5 + 0J\end{aligned}$$

Important Note

One thing to be noted is that:

$$0 \cdot J \neq 0J$$

OR

$$(0 + 0J) \cdot (0 + 1J) \neq (0 + 0J)$$

NOTE

1. **Commutativity** is intact.
2. **Distributive property** is abandoned conditionally. Its conditions are as follows:

$$X \cdot (Y + Z) \neq X \cdot Y + X \cdot Z$$

This applies if and only if all these following conditions are met:

- The multiplier (X) has a non-zero J-coefficient (b).
- The inside terms ($Y + Z$) sum to exactly absolute zero.

Otherwise, the distributive property holds and is intact.

3. **Associative property** is abandoned conditionally. Its conditions are as follows:

$$(X \cdot Y) \cdot Z \neq X \cdot (Y \cdot Z)$$

If and only if all these following conditions are met:

- An Absolute zero is involved.
- A J-number is involved.
- That J-number has a non-zero coefficient (b).

Otherwise, the Associative property holds and is intact.

The Universal Golden Formula

1st way

Let there be a function of x , such that it is in the form of $a + bJ$ for some variable x :

$$F(x) = A + Bx$$

where $\{A, B, x \in \text{J-numbers}\}$

Let:

$$A = a_1 + a_2J$$

$$B = b_1 + b_2J$$

Let x be a variable in the form of $a + bJ$.

Evaluate $F(J)$

$\Rightarrow$ substitute $x \rightarrow 0 + 1J$

$$F(0 + 1J) = [a_1 + a_2J] + [(b_1 + b_2J) \cdot (0 + 1J)]$$

Using rules of multiplication established earlier:

$$F(0 + 1J) = [a_1 + a_2J] + [b_1(0) + (b_1(1) + b_2(0) + b_2(1))J]$$

$$F(0 + 1J) = [a_1 + a_2J] + [0 + (b_1 + 0 + b_2)J]$$

Using rules of addition established earlier:

$$F(0 + 1J) = (a_1 + a_2J) + (b_1 + b_2)J$$

$$\boxed{F(0 + 1J) = a_1 + (a_2 + b_1 + b_2)J} \quad (1)$$

$F(x) = A + Bx$ is a linear J-function.

OR

Since in the function

$$F(x) = A + Bx$$

where A and B are some constants, we find the values of A and B .

$\Rightarrow$ put $x = 0$

$$F(0) = A + B(0) = A + b_2$$

Hence:

$$\boxed{A = F(0) - b_2} \quad \text{--- (1)} \quad (2)$$

$\Rightarrow$ put $x = 1$

$$F(1) = A + B(1)$$

Then:

$$B = F(1) - A$$

From Eq. (1) put $A = F(0) - b_2$:

$$B = F(1) - [F(0) - b_2]$$

Hence:

$$\boxed{B = F(1) - F(0) + b_2} \quad \text{--- (2)} \quad (3)$$

From Eq. (1) & (2) we have values of A and B now.

$\Rightarrow$ for Evaluation at J

$$F(J) = A + BJ$$

From Eq. (1) & (2):

$$\boxed{F(J) = (F(0) - b_2) + (F(1) - F(0) + b_2)J} \quad (4)$$

For ordinary functions where $b_2 = 0$

[**Important note** – All ordinary functions are subset of J-Functions], we get:

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \quad (5)$$

2nd way

Let there be a function of x , such that:

$$F(x) = \sum_{n=0}^{\infty} C_n x^n$$

$$\text{At } F(0) = C_0(1)$$

$$\text{At } F(1) = C_0 + C_1 + C_2 + C_3 + \dots$$

$$\text{Now, } F(1) - F(0) = C_1 + C_2 + C_3 + \dots (2)$$

Evaluation at $x = J$

Put $x = J$ in $F(x) = \sum_{n=0}^{\infty} C_n x^n$:

$$F(J) = C_0 + C_1 J^1 + C_2 J^2 + C_3 J^3 + C_4 J^4 + \dots$$

Since $J^n = J$ for $n > 0$:

$$F(J) = C_0 + C_1 J + C_2 J + C_3 J + C_4 J + \dots$$

$$F(J) = C_0 + (C_1 + C_2 + C_3 + C_4 + \dots)J$$

From Eq (1) and Eq (2):

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \quad (6)$$

3rd way

Let there be a function of x such that:

$$F(x) = A + Bx$$

$\Rightarrow$ A Linear J-function.

Using the Taylor Series:

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(a)}{n!} (x - a)^n$$

For our work, we center the series at $a = 0$:

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(0)}{n!} x^n$$

Using the Taylor series for our Linear J-function:

$$F(x) = A + Bx$$

Because this function is linear, the 1st derivative is constant ($F'(x) = B$) and all higher-order derivatives are zero.

Now, this becomes:

$$\boxed{F(x) = F(0) + F'(0)x} \quad \text{--- (1)} \quad (7)$$

$$F'(0) = \frac{F(x_2) - F(x_1)}{x_2 - x_1}$$

In our standard anchors $x_2 = 1$ and $x_1 = 0$, we get:

$$F'(0) = \frac{F(1) - F(0)}{1 - 0}$$

$$F'(0) = F(1) - F(0) \quad \text{--- (2)}$$

Put Eq. (2) in Eq. (1):

$$F(x) = F(0) + [F(1) - F(0)]x$$

Since at $x = J$, it becomes $x = 0 + 1J$:

$$F(0 + 1J) = F(0) + [F(1) - F(0)](0 + 1J)$$

OR

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \quad (9)$$

Sequence and Series Analysis

Sequence Analysis

Let there be a sequence a_n , such that it is treated as a function of n :

$$f(n) = a_n$$

Then, using the UGF, the infinite-th term is:

$$f(J) = f(0) + [f(1) - f(0)]J$$

$$\boxed{f(J) = a_0 + [a_1 - a_0]J} \quad (10)$$

Series Analysis (Sum of Sequences)

Let there be (sum of sequences) i.e. a series, such that, its sum up to the term n is and it is treated as a function and has a closed form:

$$S(n) = \sum_{k=0}^n a_k$$

Then, using the UGF:

$$S_J = S(0) + [S(1) - S(0)]J$$

$$S_J = a_0 + [(a_0 + a_1) - a_0]J$$

$$\boxed{S_J = a_0 + a_1J} \quad (11)$$

Absolute Zero

Absolute Zero = $0 + 0J$ [i.e., the origin]

Since $J = \frac{1}{0}$, J represents infinity.

Since J is the inverse of zero, then zero is the inverse of J :

$$\text{If } J = \frac{1}{0} = 0^{-1}$$

$$\text{Then } 0 = \frac{1}{J} = J^{-1}$$

Origins of Infinity

Since $5J$ is different from $2J$ (i.e., $5(\text{infinity})$ is different from $2(\text{infinity})$):

$$\text{Then } 5J \neq 2J$$

An infinity originated from 5 is different from an infinity originated from 2. J's (or infinities) remember their origin.

Types of Infinities and Zeros

There are different types of infinities (J's). The same is valid for zero (J^{-1}):

$$5J^{-1} \neq 2J^{-1}$$

i.e., $5(\text{zero})$ is different from $2(\text{zero})$.

Then J-inverses (or zeros) remember their origin:

$$5J^{-1} \neq 2J^{-1} \quad [\text{Information wise}]$$

A zero originated from 5 is different from a zero originated from 2.

$\therefore$ There are different types of Zeros (J-inverses)

Absolute Infinity

$$\text{Absolute Infinity} = 0 + 1J$$

NOTE: While these equations evaluate to the same Absolute Zero, or the same Absolute Infinity, the lineage, or history of how they got there, is different. *Information is preserved.*

The Generalized Universal Golden Formula

The generalized UGF is defined as:

$$\boxed{F(a + bJ) = F(a) + [F(a + b) - F(a)]J} \quad (12)$$

Binomial Expansion in J-Space

Let there be a binomial expansion such that:

$$(a + bJ)^n$$

Then it is equal to:

$$(a + bJ)^n = a^n + \left[\sum_{k=1}^n \binom{n}{k} a^{n-k} b^k \right] J \quad (13)$$

Example: $\rightarrow f(x) = x^2$

Substitute $x = a + bJ$:

$$(a + bJ)^2 = a^2 + b^2 J^2 + 2abJ$$

Since $J^n = J$ for $n > 0$, we get:

$$\boxed{(a + bJ)^2 = a^2 + (2ab + b^2)J} \quad (14)$$

Taylor Expansion for J^{-1}

Since:

$$\frac{1}{0} = J = 0^{-1}$$

Then:

$$\frac{1}{J} = 0 = J^{-1}$$

Using the Taylor expansion series, we similarly get:

$$\boxed{F(x + J^{-1}) = F(x) + \left[F'(x) + \frac{F''(x)}{2!} + \frac{F'''(x)}{3!} + \dots \right] J^{-1}} \quad (15)$$

Negative Factorials

Using the standard recurrence relation for factorials:

$$n! = n \cdot (n - 1)!$$

and the base case established in this framework:

$$(-1)! = J$$

we obtain the following sequence of values by applying the recurrence repeatedly:

$$\begin{aligned} (-2)! &= -J \\ (-3)! &= \frac{J}{2} \\ (-4)! &= -\frac{J}{6} \\ (-5)! &= \frac{J}{24} \\ (-6)! &= -\frac{J}{120} \\ &\vdots \end{aligned}$$

In general, for any positive integer n :

$$\boxed{(-n)! = \frac{(-1)^{n-1} J}{(n-1)!}} \quad (n \geq 1)$$